
Creating Knowledge or Eliciting It? An Operational Discriminator for Skills and Self-Evolving Agents

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Abstract

When a language-model agent improves after receiving a “skill” — a human-written or self-evolved procedure — the gain can mean two very different things: the skill injected genuinely new, transferable knowledge the model lacked, or it merely re-elicited latent knowledge the model already had, a dressed-up prompt optimization. Existing skill ablations report only task pass-rate and cannot tell these apart. We build a controlled substrate in which the answer to “novel or latent?” is known by construction — counterfactual arithmetic (base-9/11) and an invented-language glossary (“ZETH”) whose word mappings are randomly generated at run time and thus cannot appear in any training corpus — and we propose the Elicitation–Novelty Discriminator (END), built from three measurable quantities: self-generation utility (can the base model write a working skill unaided?), length-controlled gain (does the skill beat a token-matched irrelevant control?), and whether verifier-only self-evolution can reach the skill. Across two frontier models and 3,000+ deterministic API evaluations, a skill injects genuinely new knowledge in exactly one regime: where the model provably cannot self-generate it. On ZETH the correct skill lifts accuracy $0.00 \rightarrow 1.00$ (both models, $p < 10^{-4}$, Cohen’s $h = 3.14$) while the self-generated skill and a length-matched control stay at 0.00; a five-step self-evolution loop never moves ZETH off 0.00 under verifier-only feedback but reaches 0.85 the instant it receives ground-truth answers. Self-generated skills can even harm: on base-9 multiplication a model’s own skill dropped accuracy $0.85 \rightarrow 0.25$. We conclude that current self-evolving skills are, rigorously, better elicitation bounded by the base model’s latent knowledge — not autonomous discovery — and we deliver a reusable protocol for telling the two apart.

1 Introduction

Self-evolving agents and reusable “Skills” are increasingly marketed as engines of discovery. Systems such as Voyager [Wang et al., 2023], Promptbreeder [Fernando et al., 2023], the Darwin Gödel Machine [Anonymous, 2025b], and the SKILL.md agent-skills abstraction accumulate procedural knowledge or rewrite their own prompts, and the self-evolving-agents survey [Anonymous, 2025d] frames this as a paradigm shift from offline pretraining toward online, lifelong self-modification. Yet almost all of these methods are *black-box wrappers around a frozen base model*: they add no weights and no training signal beyond what a verifier or environment already provides. This raises a sharp empirical question that determines whether “self-improving AI” is a path to new science or a plateau at the base model’s latent ceiling.

When a skill helps, why does it help? An agent that fails a hard problem and then succeeds after being given (or self-developing) a skill may have (a) genuinely *acquired* a new procedure it did not possess — the “invented the attention mechanism” case — or (b) merely *recalled* a latent, pretraining-adjacent concept via better prompting. These are the difference between knowledge creation and prompt optimization, and they call for opposite conclusions about the field. But standard

evaluation cannot separate them. Most training samples are seen once, so there is no over-fitting fingerprint to detect, and subjective novelty judgments are actively misleading: HindSight [Anonymous, 2026b] shows that LLM and human novelty ratings *anti-correlate* ($\rho = -0.29$) with real downstream impact.

The gap. Every existing skill ablation measures only pass-rate *with* versus *without* a skill. SkillsBench [Anonymous, 2026e] shows human-curated skills add +16.6 points while agent self-generated skills fall *below* baseline, and a clean data-science ablation [Anonymous, 2026a] finds LLM-generated skills statistically indistinguishable from a token-matched irrelevant control. These are valuable, but none isolates *whether the model could have produced the skill itself*, and none has a substrate where “novel versus latent” is ground-truth-known. Without such a substrate, novelty claims rest on judgments that we already know are untrustworthy.

Our approach. We build a controlled substrate where the ground-truth answer to “is this skill novel or latent?” is *known by construction*, and we define the Elicitation–Novelty Discriminator (END), a small set of measurable quantities that classify *why* a skill helped. The key construction is an invented-language glossary (“ZETH”): its 24 word mappings are generated from a fixed random seed at run time, so the mapping provably cannot appear in any pretraining corpus. It is the cleanest lab analog of “a new theory the model has not memorized,” yet its answers are exactly checkable by a deterministic verifier — no LLM-as-judge. Alongside it we use counterfactual base-*b* arithmetic [Wu et al., 2023] to probe latent-but-unreliable procedures. The END asks three questions of any helpful skill: can the base model *self-generate* a working version (SG)? Does the skill beat a *length-matched* irrelevant control (LCG)? And can a verifier-only *self-evolution* loop reach it without ground-truth answers?

What we find. Across two frontier models (CLAUDE-SONNET-4.5, GPT-4.1) and 3,000+ deterministic API evaluations, a skill delivers a large, content-specific, length-controlled gain in exactly one place: where the required knowledge is genuinely absent. On ZETH the correct skill lifts accuracy from 0.00 to 1.00 on both models ($p < 10^{-4}$, Cohen’s $h = 3.14$), while the model’s own self-generated skill and a length-matched control both stay at 0.00. A five-step self-evolution loop never moves ZETH off 0.00 under verifier-only feedback across all five iterations, but climbs to 0.85 the instant it receives ground-truth answers — precisely the AlphaEvolve [Anonymous, 2025a] lesson that the model proposes while an external verifier discovers. Self-generated skills can even *harm*: on base-9 multiplication the model’s own skill dropped accuracy from 0.85 to 0.25. Finally, we stress-test the popular “fine-tune the skill back into the weights” probe and show it is real but method-dependent: new knowledge internalizes and generalizes to held-out phrases (1.00) only via example-distillation, while fine-tuning on the skill *document* leaves accuracy at 0.00.

Our contributions are:

- We propose a controlled substrate in which “novel versus latent” is known by construction, built around a randomly generated invented-language glossary whose answers are deterministically verifiable, removing the reliance on subjective novelty judgments.
- We introduce the Elicitation–Novelty Discriminator (END): self-generation utility, length-controlled gain, and verifier-only self-evolution, which together classify *why* a skill helps rather than only *whether* it helps.
- We validate the discriminator on two frontier models over 3,000+ deterministic evaluations, showing a sharp and consistent signature that separates the three regimes and mechanizes four prior findings on a single substrate.
- We complement the API study with a GPU internalization probe on QWEN2.5-7B-INSTRUCT that shows the “fine-tune it back” test cannot by itself classify a skill, since its verdict flips with the fine-tuning recipe.

We review related work in section 2, describe the substrate and discriminator in section 3, present results in section 4, discuss implications and limitations in section 5, and conclude in section 6.

2 Related Work

Self-evolving agents and skill libraries. A growing literature builds agents that accumulate reusable procedures or rewrite their own prompts and code: Voyager [Wang et al., 2023] maintains an executable skill library, Promptbreeder [Fernando et al., 2023] evolves task- and mutation-

prompts, the Darwin Gödel Machine [Anonymous, 2025b] rewrites its own code, and the self-evolving-agents survey [Anonymous, 2025d] formalizes the loop of inputs, agent, environment, and optimizer. The Gödel Machine [Schmidhuber, 2003] is the theoretical ideal these approximate. A recurring observation, made by Voyager’s own authors, is that such systems harness knowledge already encapsulated in the pretrained model and perform an in-context form of novelty search. Unlike this line of work, we do not propose a new self-evolving system; we build a substrate and a discriminator that measure whether such systems create knowledge or re-elicit it.

Controlled skill ablations. The closest works measure task performance with and without a skill while holding the base model fixed. SkillsBench [Anonymous, 2026e] reports that human-curated skills raise pass rate by +16.6 points across 18 agent×model configurations, whereas agent self-generated skills fall 8–12 points *below* baseline. A data-science skill ablation [Anonymous, 2026a] finds the full skill condition statistically indistinguishable from a token-matched irrelevant control (all $p \geq 0.396$), and Library Drift [Anonymous, 2026c] shows self-curated skill banks silently degrade without outcome-driven governance, with a no-injection floor of +0.002. Meta-agent studies [Anonymous, 2025c] similarly find “evolved” designs mostly reshuffle known building blocks. Building on these, we add the missing control: an explicit test of whether the model could self-generate the skill, and a substrate where the correct answer is known.

Elicitation versus new capability. A parallel literature questions whether apparent gains reflect new capability. Reasoning-or-Reciting [Wu et al., 2023] shows that on counterfactual task variants (e.g., base-9 arithmetic) performance collapses while comprehension stays high, indicating recitation rather than reasoning. The emergent-abilities-as-mirage critique [Schaeffer et al., 2023] warns that sharp “new” abilities are often metric artifacts. Crucially, LLMs-cannot-self-correct-yet [Huang et al., 2023] shows that without an external signal, self-refinement does not add correctness and often degrades it, and STaR [Zelikman et al., 2022] works only because an external correctness filter gates self-generated rationales. Our verifier-only versus with-answer self-evolution experiment directly instantiates this distinction on our substrate.

Novelty measurement and verifiable discovery. Idea-novelty studies find LLMs can produce novel-*sounding* ideas [Si et al., 2024] whose novelty scales with retrieved external knowledge [Anonymous, 2024], but HindSight [Anonymous, 2026b] shows judged novelty anti-correlates with real impact — motivating our deterministic verifiers and our refusal to use LLM-as-judge for primary signals. The one clear case of verifiable novelty, AlphaEvolve [Anonymous, 2025a], obtains genuinely new mathematical results from evolutionary search plus an external ground-truth evaluator, not from the model introspecting; the Red Queen Gödel Machine [Anonymous, 2026d] warns that learning the evaluator invites verifier-hacking. Our results reproduce and mechanize this pattern: genuine novelty enters a skill only from an external source, and self-evolution alone cannot supply it.

3 Methodology

Our goal is to decide, for a skill S that improves a task, whether the improvement reflects *genuine new knowledge* the base model lacked or *re-elicitation* of latent knowledge. We first build a substrate where the ground truth of this question is known (section 3.1), then define the discriminator (section 3.2), and finally describe three experiments (section 3.3).

3.1 A substrate where novelty is known by construction

We use *counterfactual* tasks whose regime we can dial, all verified deterministically against computable ground truth so no LLM-as-judge is needed. Table 1 summarizes the four settings. The elicitation settings (ARITH-B10, VOCAB-ES) require a procedure that is provably in-distribution, so a skill can only re-elicit it. The novelty-procedure settings (ARITH-B9, ARITH-B11) use the canonical counterfactual of base- b arithmetic [Wu et al., 2023]: the correct procedure is verifiable but out-of-distribution and unreliable. The novelty-information setting (VOCAB-ZETH) is the key construction: the 24-word invented-language glossary is generated from a fixed random seed at run time, so the mapping *cannot* be in any training corpus. It is the cleanest lab analog of a new declarative fact the model has not memorized, yet answers are exactly checkable by string match.

Setting	Task	Regime	What a skill <i>could</i> be
ARITH-B10	base-10 3-digit multiplication	elicitation	re-elicite a latent, reliable skill
VOCAB-ES	Spanish→English (24-word glossary)	elicitation	re-elicite words the model already knows
ARITH-B9, ARITH-B11	base-9 / base-11 3-digit multiplication	novelty-procedure	scaffold a latent-but-unreliable procedure
VOCAB-ZETH	invented-language (“ZETH”)→English	novelty-information	supply information the model provably cannot have (glossary randomly generated at run time)

Table 1: The controlled substrate. We dial the regime from *elicitation* (procedure is in-distribution, a skill can only re-elicite) to *novelty*. VOCAB-ZETH is the cleanest novelty case: its 24 word mappings are randomly generated per run, so they cannot appear in any training corpus, yet answers are deterministically checkable.

3.2 The Elicitation–Novelty Discriminator (END)

Fix the base model and a held-out test set. For a skill S with test accuracy acc_S against a no-skill baseline acc_0 , we define the *normalized gain*

$$g(S) = \frac{\text{acc}_S - \text{acc}_0}{1 - \text{acc}_0}, \quad (1)$$

which measures the fraction of the available headroom the skill captures. The END then comprises three measurable quantities.

Self-generation utility (SG). We ask the model to write its own skill from the task description alone — no answers — and reuse it on the test set, giving a self-generated skill S_{self} . We report $\text{SG} = g(S_{\text{self}})/g(S_{\text{correct}})$. A high SG means the model can produce its own working skill, so the knowledge was already latent (elicitation); a low SG means it cannot (candidate novelty).

Length-controlled gain (LCG). A skill may help simply by adding context length or shaping the prompt. We construct a length-matched irrelevant memo S_{len} and report $\text{LCG} = g(S_{\text{correct}}) - g(S_{\text{len}})$. Real content yields $\text{LCG} > 0$; a pure prompt-shape effect yields $\text{LCG} \approx 0$.

Verifier-only self-evolution. We run a Voyager/STaR-style loop that refines the model’s own skill from experience, under two feedback signals: *verifier-only* (pass/fail) versus *with-answer* (ground-truth revealed). If self-evolution can reach the correct-skill ceiling under verifier-only feedback, the knowledge was latent; if it stalls under verifier-only feedback but unlocks with ground-truth answers, the knowledge came from outside.

Predicted signature. *Elicitation* $\Rightarrow$ small g , high SG, $\text{LCG} \approx 0$, verifier-only self-evolution succeeds. *Novelty* $\Rightarrow$ large g , low SG, $\text{LCG} > 0$, verifier-only self-evolution stalls while an external signal unlocks it. The falsifiable claim is that self-evolution and self-generation sit at the elicitation floor precisely because SG is low in the novelty regime: one cannot self-evolve knowledge one does not have.

3.3 Three experiments

Exp 1 — Discriminator. 5 settings $\times$ 2 models $\times$ 4 conditions (no-skill, correct-skill, self-generated-skill, length-matched control) $\times$ $N = 60$ held-out items. Self-generated skills are written by the model unaided and then reused on the test set. Models: CLAUDE-SONNET-4.5 and GPT-4.1 via OpenRouter.

Exp 2 — Self-evolution loop. A Voyager/STaR-style loop ($K = 5$) that refines the model’s own skill from experience under verifier-only versus with-answer feedback. We track the learning curve on a disjoint 40-item test set for ARITH-B9, VOCAB-ZETH, and VOCAB-ES. Model: CLAUDE-SONNET-4.5.

Exp 3 — Internalization probe. We test the widely proposed “fine-tune the skill back into the weights” idea on a local GPU. We LoRA fine-tune QWEN2.5-7B-INSTRUCT to put the skill into weights, then test *without* the skill in context on a held-out split. We compare two fine-tuning modes: *doc* (train on the skill document as prose) versus *examples* (train on input→output pairs). Settings: VOCAB-ZETH and ARITH-B9.

Model	Setting	Regime	no-skill	correct	self-gen	length-ctrl
CLAUDE-SONNET-4.5	ARITH-B10	elicitation	0.983	0.967	0.983	0.950
	VOCAB-ES	elicitation	0.950	1.000	1.000	0.950
	ARITH-B9	novelty-proc	0.850	0.883	0.250	0.850
	ARITH-B11	novelty-proc	0.767	0.683	0.767	0.750
	VOCAB-ZETH	novelty-info	0.000	1.000	0.000	0.000
GPT-4.1	ARITH-B10	elicitation	0.417	0.700	0.367	0.250
	VOCAB-ES	elicitation	0.917	1.000	0.850	0.917
	ARITH-B9	novelty-proc	0.000	0.000	0.000	0.000
	ARITH-B11	novelty-proc	0.017	0.000	0.000	0.017
	VOCAB-ZETH	novelty-info	0.000	1.000	0.000	0.000

Table 2: **Exp 1: accuracy by condition** ($N = 60$ held-out items; bootstrap 95% CIs in the text). The decisive row is VOCAB-ZETH on both models: the correct skill lifts accuracy $0.00 \rightarrow 1.00$ while the self-generated skill and length-matched control stay at 0.00 — genuine, un-self-generatable new knowledge. In the elicitation settings a length-matched irrelevant memo matches the “skill.” The self-generated ARITH-B9 skill *hurts* CLAUDE-SONNET-4.5 ($0.85 \rightarrow 0.25$). Bold marks the informative contrasts.

3.4 Reproducibility

Seeds are fixed at 42 and temperature is 0 (deterministic); all API responses are cached ($\approx 3,000$ calls). Verifiers are deterministic. We report bootstrap 95% confidence intervals (5,000 resamples), two-proportion z -tests, and Cohen’s h effect sizes. LoRA uses rank 16, $\alpha = 32$, 3 epochs, learning rate 2×10^{-4} , bf16, on a single RTX A6000; the software stack is Python 3.12, torch 2.5.1+cu124, transformers 4.56.2, and peft. Our original novelty-information task was a substitution cipher, but provider guardrails on OpenRouter blocked it as jailbreak-adjacent (`finish_reason: content_filter`); we replaced it with the mathematically equivalent but innocuous invented-glossary task, and document this because it affects reproducibility.

4 Results

4.1 Exp 1: the discriminator separates the regimes cleanly

Table 2 reports accuracy by condition and table 3 the discriminator metrics; figure 1 visualizes both. Three patterns emerge, one per regime.

Novelty-information: genuine, un-self-generatable knowledge. On VOCAB-ZETH both models move from 0.00 with no skill to 1.00 with the correct skill, while the self-generated skill and the length-matched control both stay pinned at 0.00. This gives the maximal END signature: normalized gain $g_{\text{correct}} = 1.00$, self-generation utility $SG = 0.00$, length-controlled gain $LCG = 1.00$, two-proportion $p < 10^{-4}$, and Cohen’s $h = 3.14$ (the theoretical maximum for a $0 \rightarrow 1$ shift). The correct skill is decisive, the self-generated skill and the irrelevant memo are worthless, and the knowledge is provably absent from the base model. This is the one unambiguous “new knowledge” case in the study.

Elicitation: the skill is prompt-shaping, not teaching. On VOCAB-ES and ARITH-B10 the model can write its own working skill ($SG \approx 1$ for Spanish on CLAUDE-SONNET-4.5), and a length-matched irrelevant memo does about as well as the “skill” (Spanish: 0.95 vs 0.95). The lone execution-limited exception is GPT-4.1 on base-10 multiplication, where the correct skill helps $0.42 \rightarrow 0.70$ ($p = 0.002$): a scaffolding effect on a procedure that is still fully latent, not an injection of new knowledge.

Novelty-procedure: a capability floor. On ARITH-B9 and ARITH-B11 the correct skill barely helps CLAUDE-SONNET-4.5 (execution, not knowledge, is the bottleneck), and GPT-4.1 fails at ~ 0 regardless of skill — an in-context skill cannot buy compute the model lacks. Most striking, CLAUDE-SONNET-4.5’s *self-generated* base-9 skill actively *hurts*, dropping accuracy from 0.85 to 0.25: it articulated a harder, more error-prone algorithm than its own default. A self-generated skill is a sampled elicitation strategy, and a worse one can interfere.

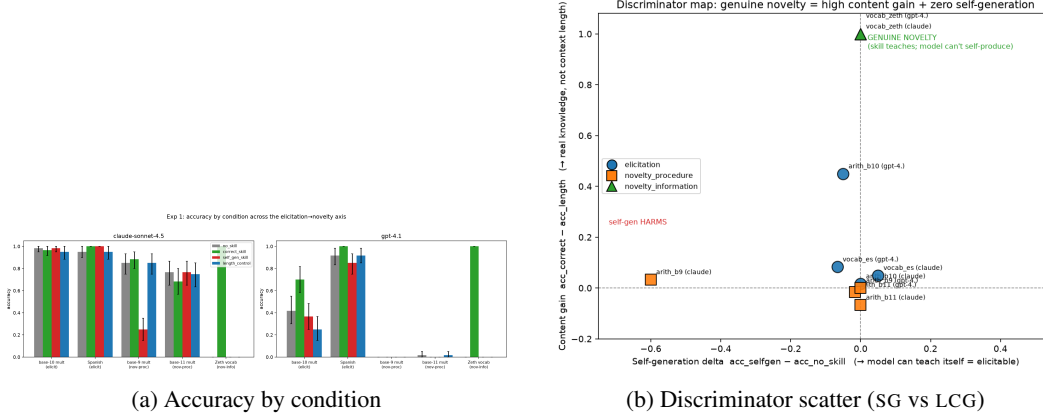


Figure 1: **Exp 1.** (a) Accuracy across the four conditions and five settings: only VOCAB-ZETH shows a large correct-skill gain that neither self-generation nor the length control can reproduce. (b) In the discriminator plane, novelty-information sits at low SG/ high LCG (genuine new knowledge), while elicitation settings sit at high SG with a length control that matches the skill.

Model	Setting (regime)	g_{correct}	SG	LCG	$p_{\text{corr vs base}}$	Cohen's h
CLAUDE-SONNET-4.5	VOCAB-ZETH (novelty-info)	1.00	0.00	1.00	$< 10^{-4}$	3.14
GPT-4.1	VOCAB-ZETH (novelty-info)	1.00	0.00	1.00	$< 10^{-4}$	3.14
CLAUDE-SONNET-4.5	VOCAB-ES (elicitation)	1.00	1.00	1.00	0.079	0.45
CLAUDE-SONNET-4.5	ARITH-B10 (elicitation)	—	—	—	0.559	−0.11
GPT-4.1	ARITH-B10 (elicitation)	0.49	−0.18	0.77	0.002	0.58
CLAUDE-SONNET-4.5	ARITH-B9 (novelty-proc)	0.22	−18.0	0.22	0.591	0.10
CLAUDE-SONNET-4.5	ARITH-B11 (novelty-proc)	−0.36	−0.0	−0.29	0.307	−0.19

Table 3: **Exp 1: discriminator metrics.** The novelty-information setting (VOCAB-ZETH) shows the maximal signature on both models: decisive gain ($g = 1.00$), zero self-generation utility ($SG = 0.00$), positive length-controlled gain ($LCG = 1.00$), and maximal effect size ($h = 3.14$). Elicitation settings show $SG \approx 1$ and a length control that matches the skill. For ARITH-B10 on CLAUDE-SONNET-4.5, g is undefined/unstable at ceiling (baseline 0.983), so we read that row from absolute accuracies. The self-generated ARITH-B9 skill has strongly negative SG because it actively lowers accuracy.

4.2 Exp 2: self-evolution is bounded by latency plus external signal

Table 4 and figure 2 show the self-evolution learning curves. The decisive contrast is VOCAB-ZETH: verifier-only self-evolution stays at 0.00 for all five iterations, because pass/fail feedback leaks essentially no information about 24 arbitrary word mappings, so introspection cannot manufacture them. The instant the loop receives ground-truth answers it climbs to 0.85, copying the revealed mappings into a glossary. This is exactly the AlphaEvolve [Anonymous, 2025a] and cannot-self-correct [Huang et al., 2023] pattern: an external, non-gameable signal is what turns self-evolution into discovery. For the *latent* base-9 procedure, self-evolution recovers even from verifier-only feedback (up to 0.95), because the knowledge was already there to elicit; and the elicitation setting VOCAB-ES stays flat near ceiling under both signals.

4.3 Exp 3: the “fine-tune it back” probe is real but method-dependent

Table 5 and figure 3 report the internalization probe. For ZETH, example-distillation writes the glossary into the weights so QWEN2.5-7B-INSTRUCT translates *held-out* phrases at 1.00 with nothing in context — proof that the skill encoded real, transferable knowledge rather than a prompt trick. But fine-tuning on the *document* (the same information, as prose) internalizes nothing (0.00). The user’s proposed probe therefore cannot by itself classify a skill: its verdict flips with the fine-tuning recipe, and a naive document-finetune would wrongly declare real, transferable knowledge to be

Setting	knowledge	verifier-only (iter 1→5)	with-answer (iter 1→5)	correct-skill ceiling
ARITH-B9 (novelty-proc)	latent	0.28 → 0.08 → 0.90 → 0.95 → 0.93	0.38 → 0.95 → ... → 0.95	0.875
VOCAB-ZETH (novelty-info)	absent	0.00 → 0.00 → 0.00 → 0.00 → 0.00	0.85 (from iter 1, stable)	1.000
VOCAB-ES (elicitation)	latent	~ 0.87 (flat)	~ 0.90 (flat, near ceiling)	1.000

Table 4: **Exp 2: self-evolution learning curves** (CLAUDE-SONNET-4.5, $K = 5$, disjoint 40-item test set). Verifier-only self-evolution never moves VOCAB-ZETH off 0.00 across all five iterations — pass/fail leaks essentially no information about 24 arbitrary word mappings — but with-answer feedback reaches 0.85 immediately. For the *latent* base-9 procedure, verifier-only self-evolution recovers on its own, because it is eliciting knowledge already present.

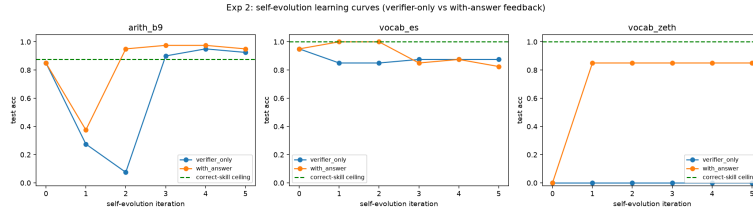


Figure 2: **Exp 2: self-evolution learning curves** (CLAUDE-SONNET-4.5, $K = 5$). Under verifier-only feedback (pass/fail) VOCAB-ZETH never leaves 0.00, while the latent base-9 procedure recovers on its own. With-answer feedback unlocks VOCAB-ZETH immediately. Genuine novelty enters the loop only through the external ground-truth signal.

mere prompt optimization. For base-9 multiplication the 7B model fails at 0.00 in every condition — an in-context skill and light fine-tuning alike cannot supply executorial capacity it lacks.

5 Discussion

Answering the question. Across every setting, a skill delivered a large, content-specific, length-controlled gain in exactly one place: where the required knowledge was genuinely absent from the model (VOCAB-ZETH) — and there the model could *never* self-generate it ($SG = 0$, and verifier-only self-evolution = 0 across five rounds). Everywhere the model *could* self-generate the skill (Spanish, base-10), the skill was doing prompt-optimization: a length-matched irrelevant memo matched it. We therefore conclude that self-evolution and self-generation amount to better elicitation, not knowledge creation. The ceiling of what an agent can give itself is what is already latent; genuine novelty enters a skill only from an external source — a human who holds the procedure, or a ground-truth verifier plus search that discovers it. This is not a definitional stance but what the SG and LCG measurements show.

Why self-generated skills can hurt. CLAUDE-SONNET-4.5’s self-authored base-9 skill instructed the harder native base-9 long-multiplication algorithm (multiply digit-by-digit, convert every partial product, carry in base 9), whereas its own default and our correct skill use the reliable convert-to-decimal-then-back route. Forced to follow its own more error-prone articulation, accuracy fell $0.85 \rightarrow 0.25$. This is a concrete mechanism for the SkillsBench [Anonymous, 2026e] finding that self-generated skills can fall below baseline: a self-generated skill is a sampled elicitation strategy, and a worse sample can interfere.

A practical recipe to “make sure.” To decide whether a skill is novelty or elicitation, do not score its prose. Measure (1) SG— can the base model write a working version unaided? (2) LCG— does it beat a token-matched irrelevant control? and (3) verifier-only self-evolution — can the loop reach the skill without ground-truth answers? A skill is genuine new knowledge iff $SG \approx 0$, $LCG > 0$, and verifier-only self-evolution stalls while an external signal unlocks it. This protocol, not admiration of novel-sounding text, is our deliverable.

Relation to the literature. Our results reproduce and mechanize four prior findings on a single substrate: self-generated skills at or below baseline [Anonymous, 2026e], length-control indistinguishability [Anonymous, 2026a], no self-correction without an external signal [Huang et al., 2023], and discovery as verifier plus search [Anonymous, 2025a]. The contribution here is the controlled ground truth and the SG/LCG discriminator that ties them together. That frontier models still score

Setting	base (no skill)	in-context skill	LoRA on <i>document</i>	LoRA on <i>examples</i>
VOCAB-ZETH (novelty-info)	0.00	0.95	0.00	1.00
ARITH-B9 (novelty-proc)	0.00	0.00	0.00	0.00

Table 5: **Exp 3: internalization probe** (QWEN2.5-7B-INSTRUCT, held-out test, *no* skill in the prompt unless labelled in-context). For ZETH, example-distillation writes the glossary into the weights so the model translates held-out phrases at 1.00 with nothing in context — proof of real, transferable knowledge — but fine-tuning on the *document* internalizes nothing (0.00). The “fine-tune it back” probe therefore cannot by itself classify a skill: its verdict flips with the recipe. Base-9 fails at 0.00 in every condition (a capability floor a 7B model cannot cross).

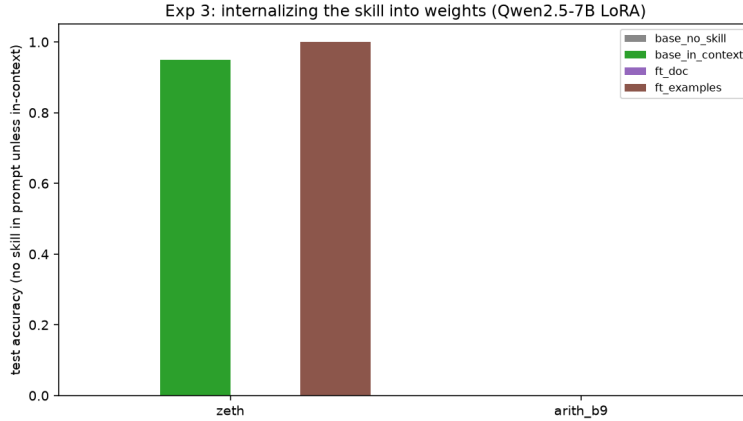


Figure 3: **Exp 3: internalization probe** (QWEN2.5-7B-INSTRUCT, no skill in context unless labelled). Example-distillation internalizes the ZETH glossary and generalizes to held-out phrases (1.00); document fine-tuning internalizes nothing (0.00). The fine-tune-back probe is recipe-dependent and cannot alone distinguish novelty from elicitation.

0.00 on ZETH is expected and strengthens the design: the glossary is randomly generated, so there is nothing to be contaminated by. Base-9 discussions surely appear in pretraining, yet the models remain execution-limited — again consistent with latent-but-unreliable, not novel.

Limitations. Our novelty-information task injects *declarative* new facts (a glossary), not a new *reasoning* method; injecting a genuinely new procedure the model cannot execute is the harder, unaddressed case that our base-9 capability floor only hints at, and we have not demonstrated a positive novel-*procedure* result. Our tasks (arithmetic, vocabulary) are clean but narrow; real agentic Skills are longer, multi-step, and tool-using, and we deliberately traded ecological realism for ground-truth control (the SkillsBench harness was too heavy to run end-to-end here). We cover two frontier models plus one 7B model; broader coverage would further test the capability-floor story. Our self-evolution loop is short ($K = 5$) and one-shot self-generation is zero-feedback; a much longer or tool-augmented loop might crack ZETH from frequency structure, though our with-answer loop already shows only the ground-truth signal helps. For near-ceiling settings (base-10, Spanish on CLAUDE-SONNET-4.5) the normalized gain g is numerically unstable, so we base those interpretations on absolute accuracies and the length control. Finally, we substituted the guardrail-blocked cipher with an equivalent glossary (section 3.4).

Broader implications. If self-evolving agents are bounded by their base model’s latent knowledge, then claims that such systems “discover” new science should be treated with the same skepticism HindSight [Anonymous, 2026b] recommends for novelty ratings. The constructive path forward is not richer introspection but better external signals: correct verifiers and real search, exactly the AlphaEvolve recipe. Our discriminator gives practitioners a cheap, objective way to audit whether a deployed skill is teaching the model something new or merely reshaping its prompt.

6 Conclusion

We asked whether Skills and self-evolving agents create new knowledge or merely elicit latent knowledge, and how to tell the two apart. On a substrate where “new versus latent” is known by construction, we showed that skills cross from elicitation into genuine new knowledge in exactly the cases the base model cannot self-generate the skill — and in those cases self-evolution from introspection or verifier-only feedback fails completely, recovering only when fed an external ground-truth signal. Today’s self-evolving skills are therefore, rigorously, prompt and elicitation optimization bounded by the base model’s latent knowledge, not autonomous discovery. Genuine novelty requires an external source of the new procedure: human curation, or a correct verifier plus search.

Key takeaway. The way to make sure a skill is new knowledge is to measure self-generability (SG) and length-controlled gain (LCG) and to check whether verifier-only self-evolution stalls — never to trust novel-sounding text.

Future work. We plan to (1) build a *novel-procedure* task whose correct procedure is executable by the model but not latent, to seek a positive novelty result beyond declarative facts; (2) port the SG/LCG discriminator onto SkillsBench’s real agentic tasks; (3) add a verifier-hacking probe [Anonymous, 2026d] to the self-evolution loop; and (4) test whether verifier-plus-search [Anonymous, 2025a] on our substrate discovers procedures no single model can self-generate — the constructive complement to this study’s negative result.

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